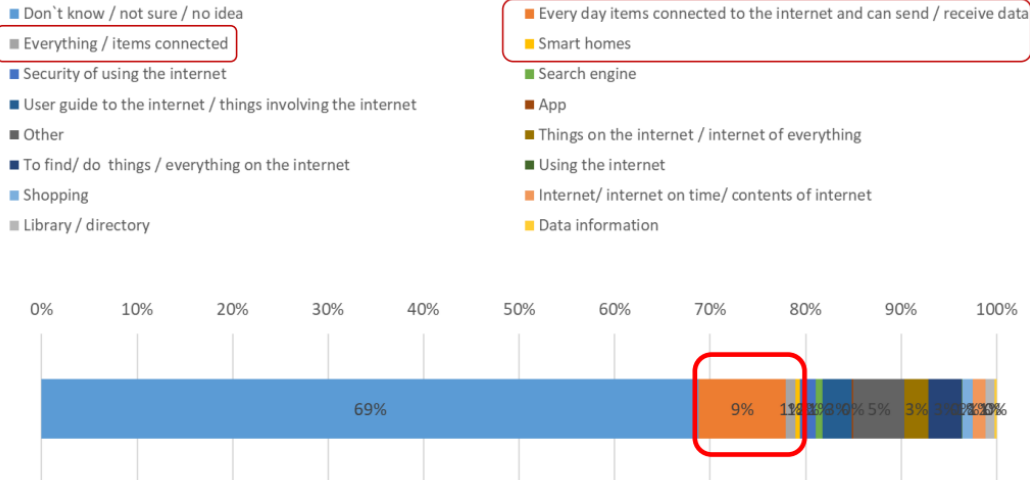


09_ INTERNET OF THINGS

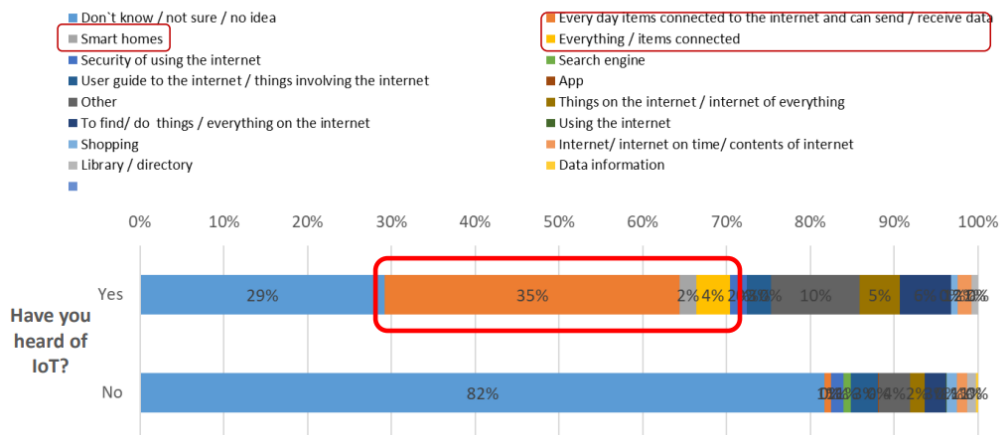
Sondaggi fatto qualche anno fa:

What is the IOT?



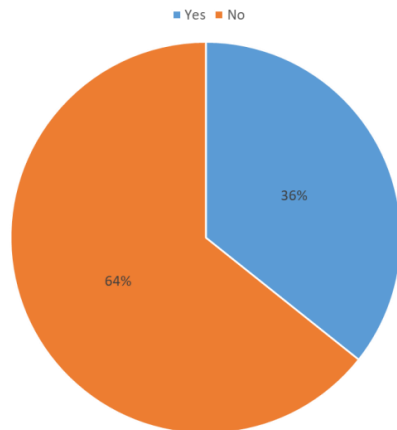
Taking in a couple of other descriptions which just about make sense, we hit around 11%.

A whole IoT of headlines

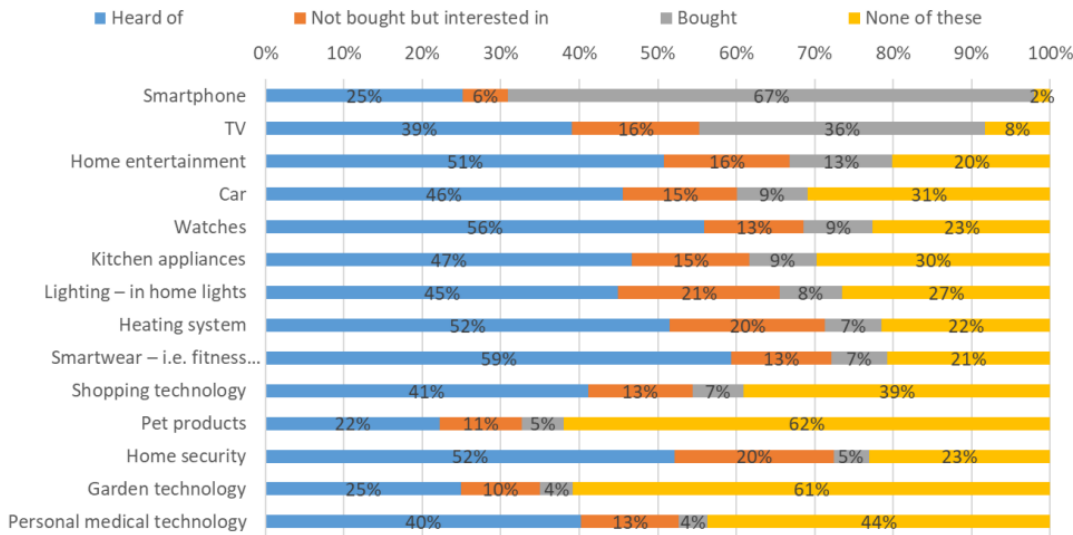


- Whilst tech writers are easing off the subject, there is a whole lot of consumer noise around the subject.
- Even those who claim to have already heard of IoT are likely to either get it wrong, or be blinded by the question.

Have you bought any items that are 'smart'?



In realtà il 100% li aveva comprati (ad esempio smartphone).



But there is plenty of headroom in the market.

IoT: latest development in the long and continuing revolution of computing and communications. The term refers to:

- expanding interconnection of smart devices
- ranging from appliances to tiny sensors
- Interconnection of billions of industrial and personal objects, usually through cloud systems

Devices in IoT: Personal computers, Smartphones,

Da quando abbiamo iniziato a parlare di IoT abbiamo introdotto dispositivi embedded con risorse limitate (Low-bandwidth, low-repetition data capture, low-bandwidth data-usage).

Radio-frequency identification (RFID) was seen as a prerequisite for the IoT at the start. If all objects and people in daily life were equipped with identifiers, computers could manage and inventory them.

Besides using RFID, the tagging of things may be achieved through such technologies as Near Field Communication, barcodes, QR codes, bluetooth, and digital watermarking.

Passaggi dell'evoluzione di IoT:

1. Information technology (IT): PCs, servers, routers, firewalls, and so on.
Bought as IT devices by enterprise IT people, primarily using wired connectivity.
2. Operational technology (OT): machines/appliances with embedded IT built by non-IT companies, such as medical machinery, SCADA (supervisory control and data acquisition), process control, and kiosks.
Bought as appliances by enterprise OT people, primarily using wired connectivity.
3. Personal technology: smartphones, tablets, and eBook readers bought as IT devices by consumers (employees); exclusively using wireless connectivity and often multiple forms of wireless connectivity.
4. Sensor/actuator technology: single-purpose devices bought by consumers, IT, and OT people; exclusively using wireless connectivity, generally of a single form, as part of larger systems.

Components:

- **Sensor:** measures some parameter of a physical, chemical, or biological entity and delivers an electronic signal proportional to the observed characteristic, either in the form of an analog voltage level or a digital signal.
In both cases, the sensor output is typically input to a microcontroller or other management element.
- **Actuator:** receives an electronic signal from a controller and responds by interacting with its environment to produce an effect on some parameter of a physical, chemical, or biological entity.
Ad esempio in un profumatore d'ambiente, ricevuto il segnale che la qualità dell'aria è pessima, è quello che spruzza il profumo.
- **Microcontroller:** the "smart" in a smart device is provided by a deeply embedded microcontroller.
- **Transceiver:** contains the electronics needed to transmit and receive data.
Most IoT devices contain a wireless transceiver, capable of communication using Wi-Fi, ZigBee, or some other wireless scheme.
- **Radio-Frequency Identification (RFID):** Technology that uses radio waves to identify items, is increasingly becoming an enabling technology for IoT.
The main elements of an RFID system are tags and readers. RFID tags are small programmable devices used for object, animal, and human tracking. They come in a variety of shapes, sizes, functionalities, and costs.

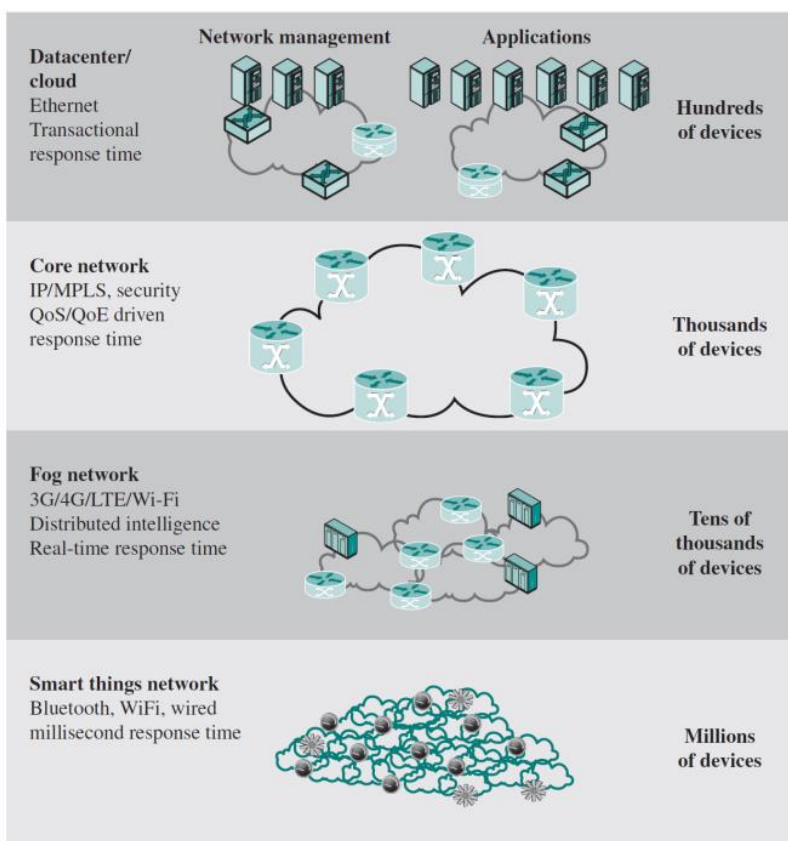
Structure of IoT: the IoT can be viewed as a gigantic network consisting of networks of devices and computers connected through a series of intermediate technologies where numerous technologies like RFIDs, wireless connections may act as enablers of this connectivity.

Tagging Things: Real-time item traceability and addressability by RFIDs.

Feeling Things: Sensors act as primary devices to collect data from the environment.

Shrinking Things: Miniaturization and Nanotechnology has provoked the ability of smaller things to interact and connect within the “things” or “smart devices”.

Thinking Things: Embedded intelligence in devices through sensors has formed the network connection to the Internet. It can make the “things” realizing the intelligent control.



Spesso quando si parla di IoT spesso si intende dispositivi embedded.

Constrained devices:

- very limited resources including limited RAM (kbytes) and ROM
- low-power requirements (batteries)
- no memory management unit
- limited processor performance

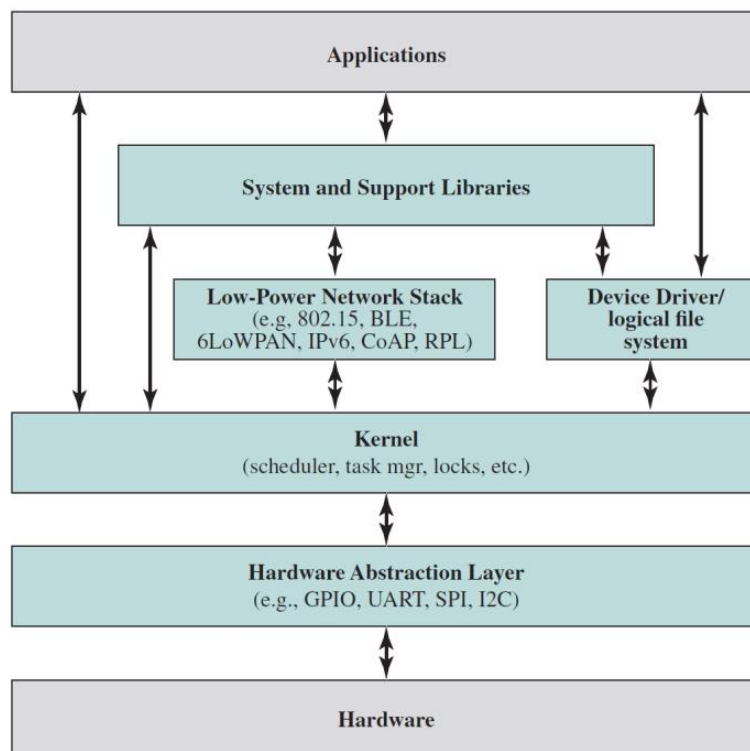
Abbiamo bisogno di OS appositi per sistemi embedded (i più usati sono TinyOS, RIOT, μ Clinux).

IoT OS requirements:

- **Small memory footprint:** few available memory, we need to place both OS and applications. Need for libraries optimized in terms of both size and performance, and space-efficient data structures.

- **Support for heterogeneous hardware:** for the largest systems, such as servers, PCs, and laptops, the Intel x86 processor architecture dominates.
For smaller systems, such as smartphones and a number of classes of IoT devices, the ARM architecture dominates.
Constrained devices are based on various microcontroller architectures and families, especially 8-bit and 16-bit processors.
A wide variety of communications technologies are also deployed on constrained devices.
- **Network connectivity:** IEEE 802.15.4 [low-rate wireless personal area network (WPAN)] (come ZigBee).
Bluetooth Low Energy (BLE), 6LoWPAN (IPv6 over Low-power Wireless Personal Area Networks), CoAP (Constrained Application Protocol), RPL (Routing Protocol for Low power and Lossy Networks), LoRaWAN (low-power Long-Range network technology).
- **Energy efficiency:** need to work for years with a single battery. Making the processor as energy efficient as possible, wireless transmission schemes that minimize energy consumption, OS must provide energy consumption facilities (provide them to higher level, use them).
- **Real-time capabilities:** several devices must support real-time applications (Surveillance, Smart cars). OS must fulfill timely execution requirements.
- **Security:** at application layer (Authorization and Authentication, Data confidentiality, Anti-virus, Privacy), at network layer (Authorization and Authentication, Data confidentiality, Data integrity protection), at device layer (Authorization and Authentication, Device integrity validation, Access control).

Typical structure for IoT OS:

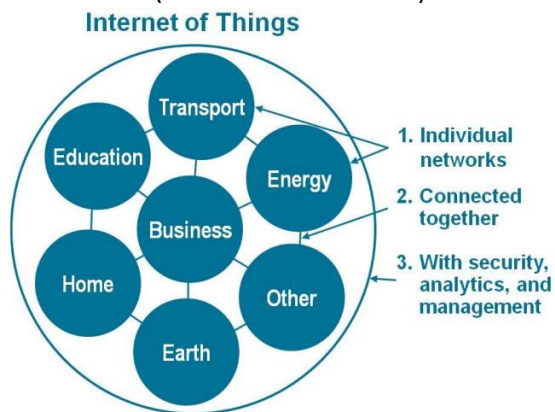


Caratteristiche degli OS più usati:

	μ Clinux	TinyOS	RIOT
Minimum RAM	< 32 MB	< 1 kB	~1.5 kB
Minimum ROM	< 2 MB	< 4 kB	~5 kB
C Support	✓	✗	✓
C++ Support	✓	✗	✓
Multithreading	✓	○	✓
Microcontrollers without MMU	✓	✓	✓
Modularity	○	✗	✓
Real time	○	✗	✓

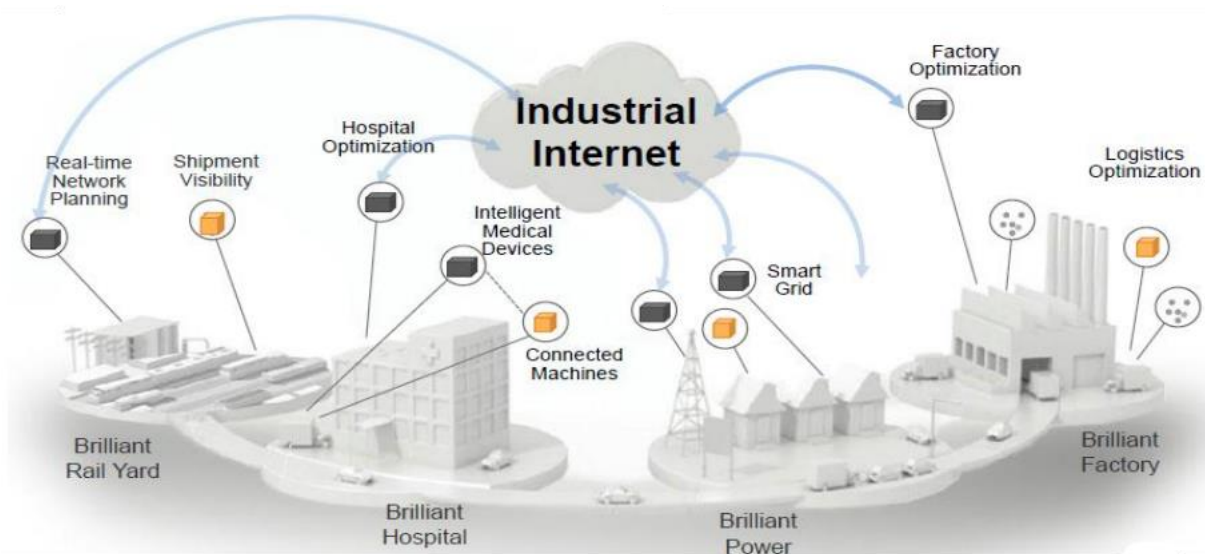
✓ = full support
○ = partial support
✗ = no support

IoT è una rete di reti (Network of Networks):

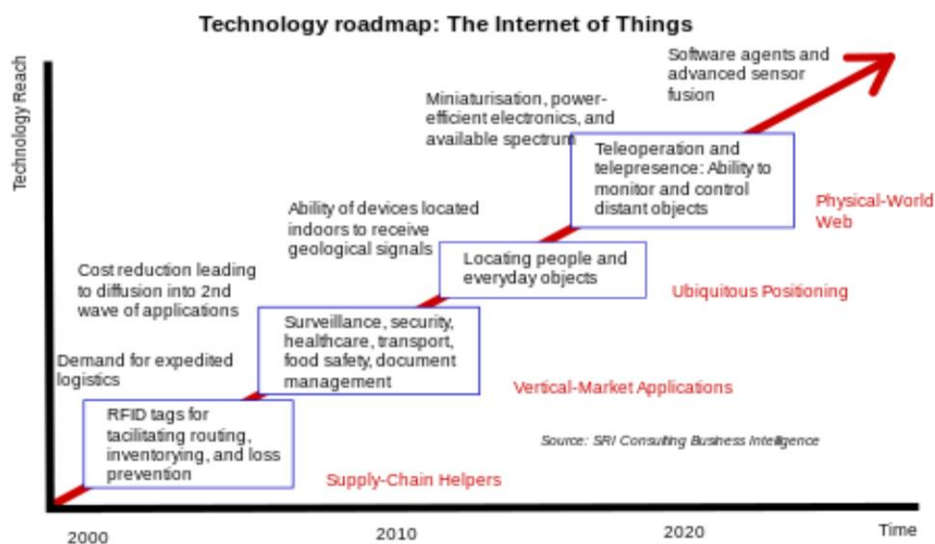
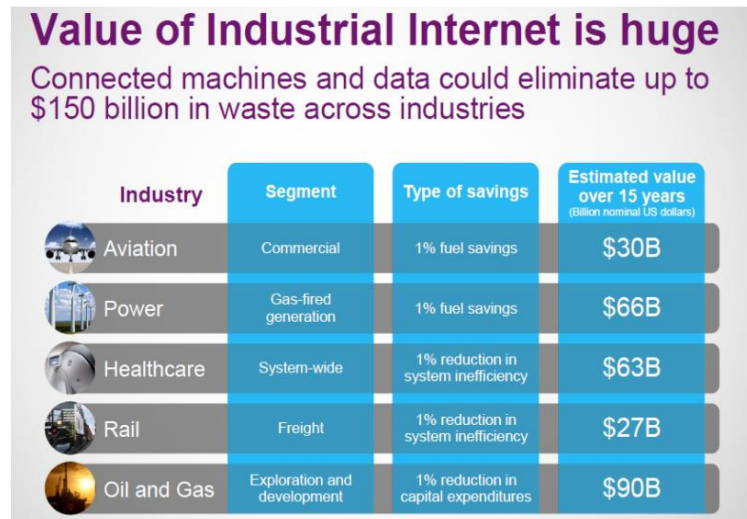


These networks connected with added security, analytics, and management capabilities. This will allow IoT to become even more powerful in what it can help people achieve.

Quando si hanno miliardi di macchine connesse:



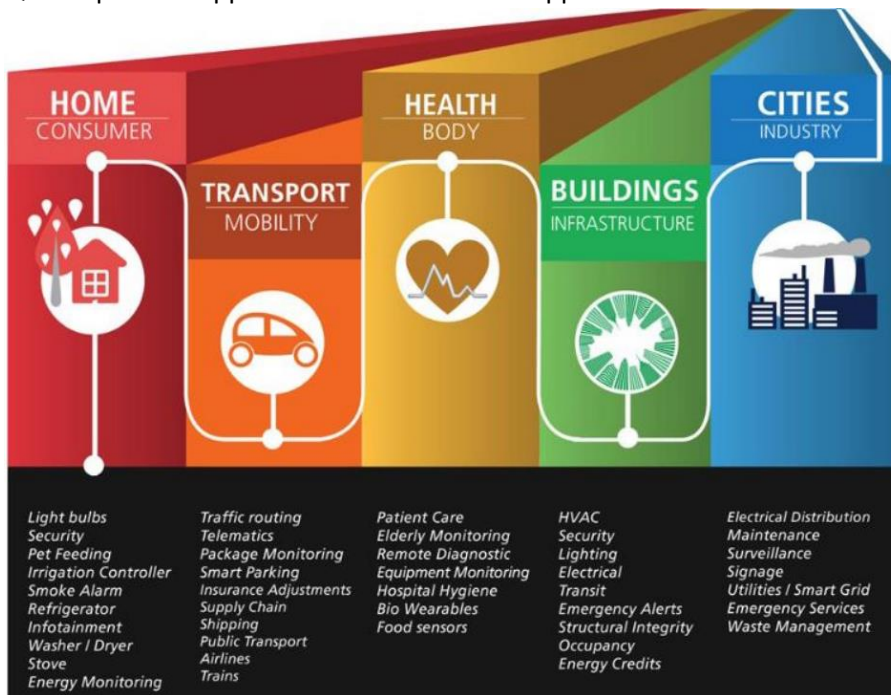
Potential of IoT:



Alcune possibili applicazioni di IoT:

- Building and Home automation
- Manufacturing
- Medical and Healthcare systems
- Media
- Environmental monitoring
- Infrastructure management
- Energy management
- Transportation
- Better quality of life for elderly
- ...

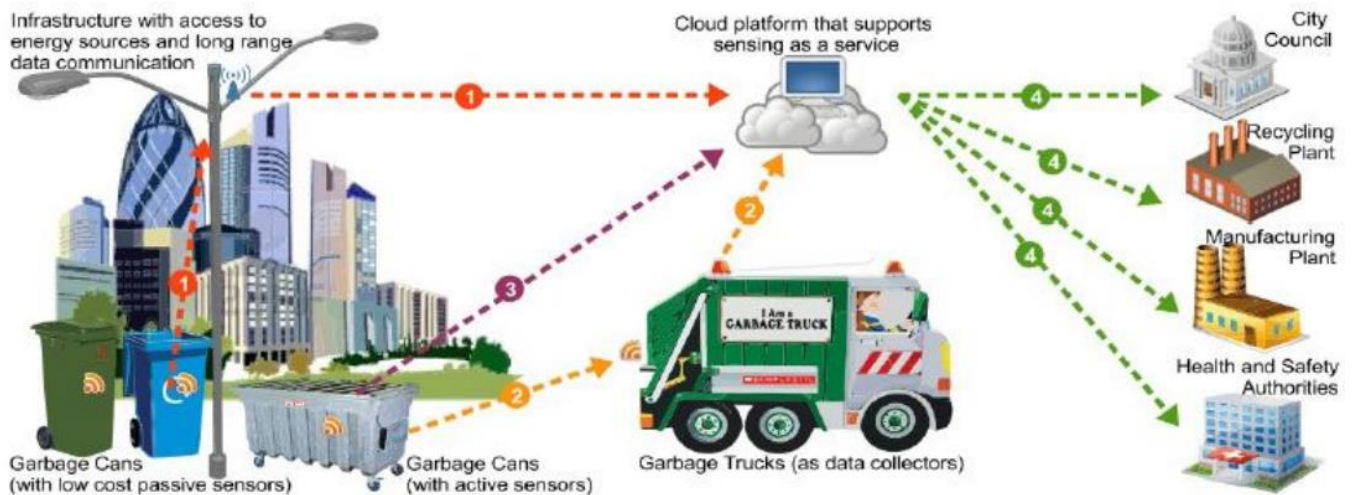
Queste possibili applicazioni sono in diverse applicazioni:



es. smart parking



Efficient waste management in smart cities supported by the sensing-as-a-service:

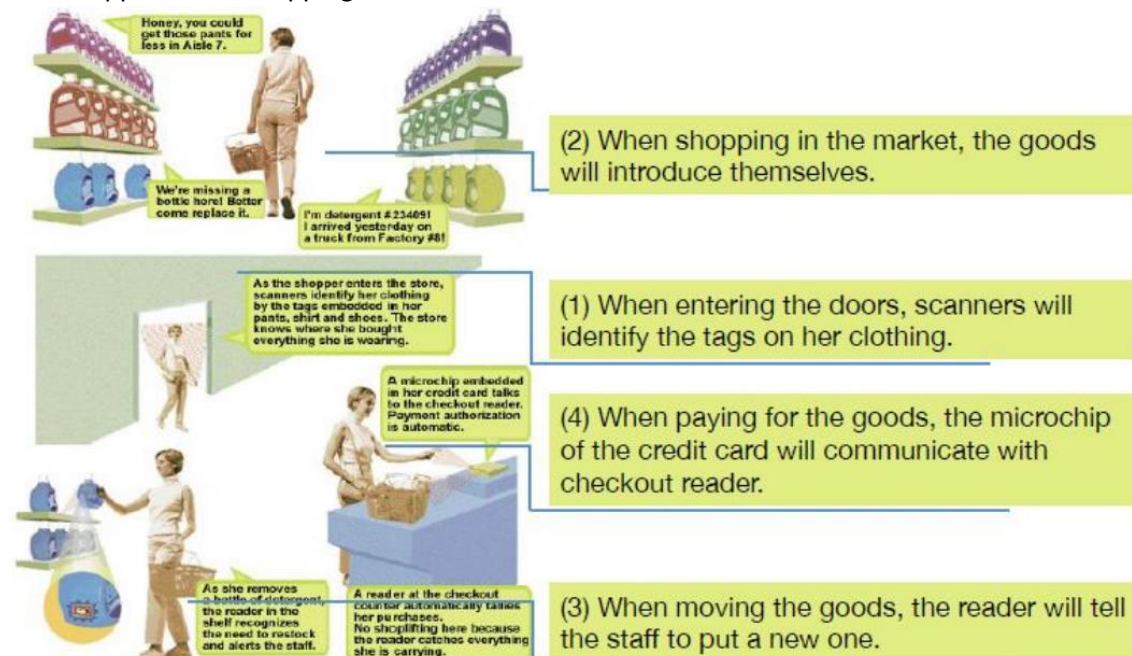


Immaginiamo un “camion” che prende le informazioni e sa dove portarle, una volta che i cestini sono quasi pieni.

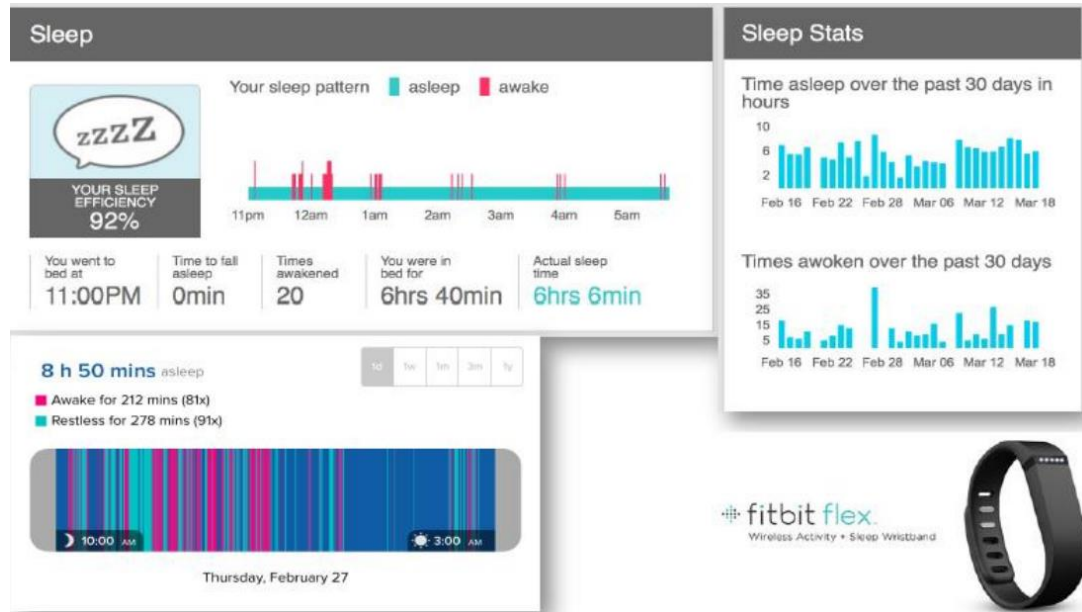
es. mucche

In the world of IoT, even the cows will be connected and monitored. Sensors are implanted in the ears of cattle. This allows farmers to monitor cows' health and track their movements, ensuring a healthier, more plentiful supply of milk and meat for people to consume. On average, each cow generates about 200 MB of information per year.

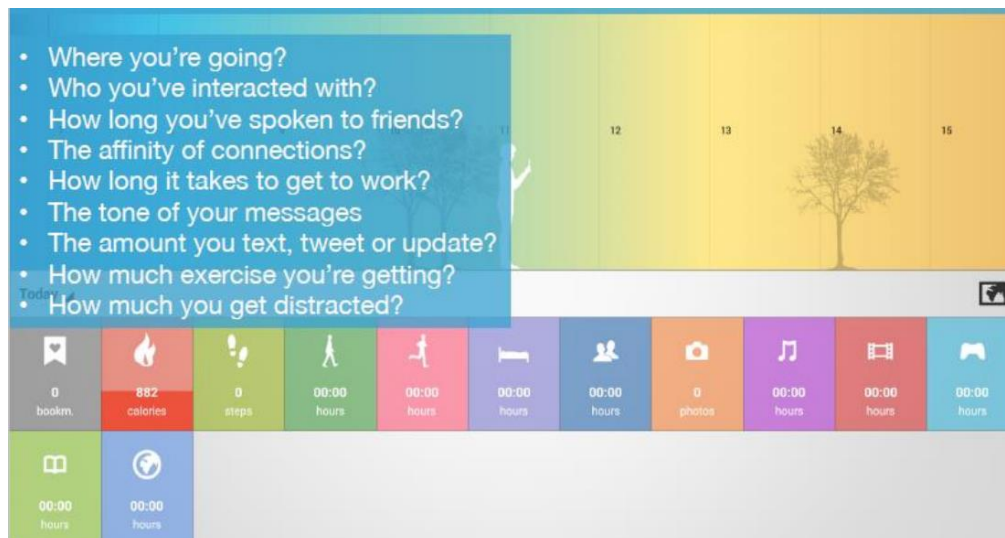
es. scenario di applicazione shopping



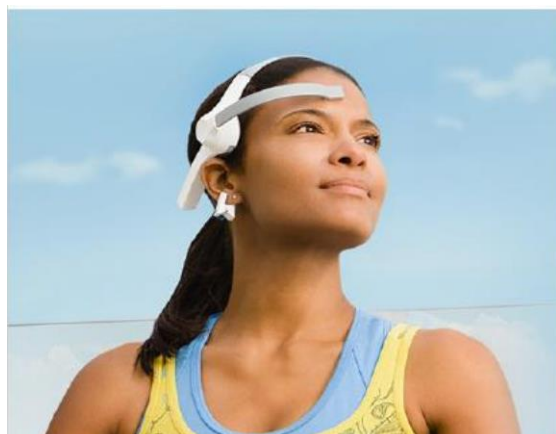
es. how well do I sleep?



es.



es. Thought Controlled Computing



The flagship product, MindWave, is a headset that can log into your computer using just your thoughts. Researchers recently used the EEG headset to develop a toy car that can be driven forward with thought.

NeuroSky's smart sensors can also track your heart rate and other bodily metrics and can be embedded in the next generation of wearable devices.

"We make it possible for millions of consumers to capture and quantify critical health and wellness data," Yang (CEO of Softbank) said. Softbank is the funder.

Tutte queste applicazioni generano tantissimi dati, ai quali dobbiamo essere in grado di dare senso. It doesn't matter how much data you have if you can't make sense of it. Big Data is not magic.

Criticisms and Controversies of IoT: there are doubts about the promises of the ubiquitous computing revolution, in the areas as:

- Privacy
- Security
- Autonomy and Control
- Social control
- Political manipulation
- Design
- Environmental impact
- Influences human moral decision making

To summarise: IoT is only the tip of an iceberg